

A.2 Work, Energy and Power

Past Paper Answers

These are past paper answers selected from past papers (2016 - 2022) that relate to the topic A.3 Work, Energy and Power.

The associated questions are in the separate accompanying document

Answer 1

B

Answer 2

a	i	$«E_{el} = \frac{1}{2}mv^2 + mgh$ OR $«E_{el} = E_p + E_k \checkmark$ $«E_{el} = \frac{1}{2} \times 55 \times 0.90^2 + 55 \times 9.8 \times 1.2$ OR $669 \text{ J } \checkmark$ $«E_{el} = 669 \approx 670 \text{ J}»$	Award [1 max] for use of $g = 10 \text{ N kg}^{-1}$, gives 682 J.	2
a	ii	$\frac{1}{2} \times 55 \times v^2 = 670 \text{ J } \checkmark$ $v = \sqrt{\frac{2 \times 670}{55}} = 4.9 \text{ m s}^{-1} \checkmark$	If 682 J used, answer is 5.0 m s^{-1}.	2

Answer 3

D

Answer 4

$\frac{1}{2}v^2 = 0.24gh \checkmark$ $v = 11.9 \text{ «ms}^{-1}\text{» } \checkmark$	Award GPE lost $= 65 \times 9.81 \times 30 = \text{«19130 J»}.$ Must see the 11.9 value for MP2, not simply 12. Allow $g = 9.8 \text{ ms}^{-2}$.	2
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Answer 5

B

Answer 6

A

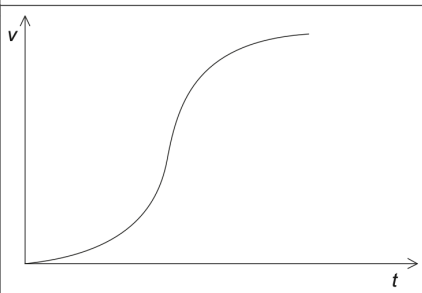
Answer 7

D

Answer 8

D

Answer 9

a	links 0.84 to Δp ✓ $v = \frac{0.84}{5.8 \times 10^{-2}} \Rightarrow 14.5 \text{ «ms}^{-1}\text{»} \checkmark$	Award [2] for bald correct answer	2
b	use of $\Delta t = (28 - 12) \times 10^{-3} \Rightarrow 16 \times 10^{-3} \text{ «s»} \checkmark$ $\bar{F} = \frac{\Delta p}{\Delta t} = \frac{0.84}{16 \times 10^{-3}} \text{ OR } 53 \text{ «N»} \checkmark$	Accept a time interval from 14 to 16 ms Allow ECF from incorrect time interval	2
c	$E_k = \frac{1}{2} \times 5.8 \times 10^{-2} \times 14.5^2 \checkmark$ $E_k = W \checkmark$ $s = \frac{W}{F} = \frac{\frac{1}{2} \times 5.8 \times 10^{-2} \times 14.5^2}{53} \Rightarrow 0.12 \text{ «m»} \checkmark$	Allow ECF from (a) and (b) Allow ECF from MP1 Award [2] max for a calculation without reference to work done, eg: average velocity $\times$ time	3
d	 graph must show increasing speed from an initial of zero all the time ✓ overall correct curvature ✓		2

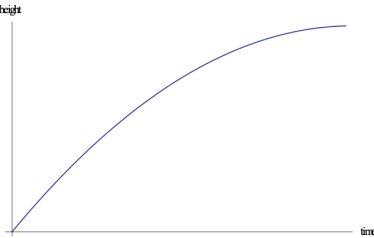
Answer 10

B

Answer 11

D

Answer 12

a		$H = \frac{1}{2}gt^2 \Rightarrow 4.9 \text{ «m»} \checkmark$	Accept other methods as area from graph, alternative kinematics equations or conservation of mechanical energy. Award [1] for a bald correct answer in the range 4.9 - 5.1 Award [0] if time used is different than 1.0 s	1
b	i	M at 1.6 s ✓		1
b	ii	«g» 9.80 «ms ⁻² » ✓	Accept 9.81, 10 or a plain “g” Ignore sign if provided.	1
b	iii	 concave down parabola as shown «with non-zero initial slope and zero final slope» ✓	Award [1] mark if curve starts from a positive time value. Award [0] if the final slope is negative.	1
c		« loss of KE is $\frac{1}{2} \times 0.25 \times (9.8^2 - 5^2) \Rightarrow 8.9 \text{ «J»} \checkmark$	Award [1] mark for an answer in the range 8.7 - 9.5	1
d	i	$\Delta p = 0.250 \times (9.8 + 5.0) \checkmark$ $F_{\text{net}} = \frac{\Delta p}{\Delta t} = \frac{3.7}{0.1} \Rightarrow 37 \text{ «N»} \checkmark$ $N = 37 + 0.250 \times 9.8 = 39.5 \text{ «N»} \checkmark$	Allow ECF for MP2 and MP3	3
d	ii	there is an external force acting on the ball OR some momentum is transferred to the floor ✓	Allow references to impulse instead of force. Do not award references to energy.	1

Answer 13

B

Answer 14

B

Answer 15

A

Answer 16

D

Answer 17

A

Answer 18

a		g OR 9.81 m s^{-2} OR acceleration of gravity/due to free fall ✓	Accept 10 m s^{-2} . Ignore sign. Do not accept bald "gravity". Accept answer that indicates tangent of the graph at time $t=0$.	1
b		Identification of air resistance/drag force «acting upwards» ✓ «that» increases with speed ✓ «until» weight and air resistance cancel out OR net force/acceleration becomes zero ✓	A statement as "air resistance increases with speed" scores MP1 and MP2 .	3
c	i	«loss in» $GPE = 3.4 \times 10^{-5} \times 9.81 \times 21 = 7.0 \times 10^{-3} \text{ J}$ ✓ OR «gain in» $KE = 0.5 \times 3.4 \times 10^{-5} \times 9.0^2 = 1.4 \times 10^{-3} \text{ J}$ ✓ energy transferred to air « $= 7.0 \times 10^{-3} - 1.4 \times 10^{-3} = 5.6 \times 10^{-3} \text{ J}$ » ✓ any calculated answer to 2 sf ✓	Allow [1] through the use of kinematics assuming constant acceleration. Allow ECF from MP1	3
c	ii	«gravitational» potential energy «of the raindrop» into thermal/internal energy «of the air» ✓	Accept heat for thermal energy Accept into kinetic energy of air particles Ignore sound energy	1

Answer 19

power received = $240 \times 25000 = \text{«}6.0 \text{ MW}\text{»} \checkmark$ efficiency $\text{«} = \frac{1.6}{6.0} \text{»} = 0.27 / 27\% \checkmark$		2
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Answer 20

specific energy available = $\text{«} gh = \text{»} 9.81 \times 270 \text{ «} = 2650 \text{ J kg}^{-1} \text{»}$ OR $mgh = \frac{1}{2}mv^2$ OR $v^2 = 2gh \checkmark$ $v = 73 \text{ «ms}^{-1}\text{»} \checkmark$	<i>Do not allow 72 as round from 72.8</i>	2
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